Addition zweier Sinusfunktionen gleicher Frequenz

$$\omega_1 = \omega_2 = \omega$$

$$\hat{u}_1 \neq \hat{u}_2 \;, \quad \varphi_1 \neq \varphi_2$$

$$u_1(t) = \hat{u}_1 \cdot \cos(\omega t + \varphi_1)$$

$$u_2(t) = \hat{u}_2 \cdot \cos(\omega t + \varphi_2)$$

Lösung:

$$u(t) = \hat{u} \cdot \cos(\omega t + \varphi)$$

Es gilt: $$\hat{u} = \sqrt{(\hat{u}_1 \cos\varphi_1 + \hat{u}_2 \cos\varphi_2)^2 + (\hat{u}_1 \sin\varphi_1 + \hat{u}_2 \sin\varphi_2)^2}$$

$$\varphi = \arctan\left(\frac{\hat{u}_1 \sin\varphi_1 + \hat{u}_2 \sin\varphi_2}{\hat{u}_1 \cos\varphi_1 + \hat{u}_2 \cos\varphi_2}\right)$$

Beispiel:

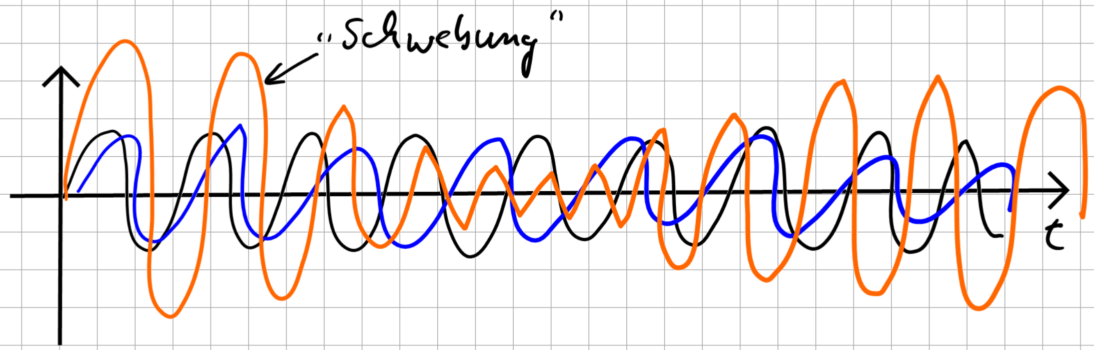
$$u_1(t) = \underbrace{1\,V}_{\hat{u}_1} \cdot \cos(\omega t \underbrace{- 10^\circ}_{\varphi_1})$$

$$u_2(t) = \underbrace{2\,V}_{\hat{u}_2} \cdot \cos(\omega t \underbrace{+ 50^\circ}_{\varphi_2})$$

$$\Rightarrow \quad u(t) = u_1(t) + u_2(t)$$

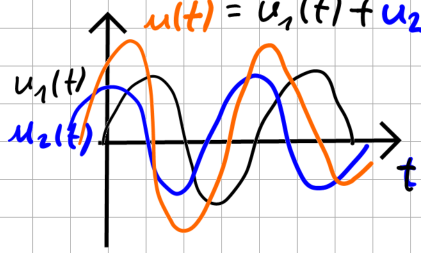
$$= \underline{\hat{u}} \cdot \cos(\omega t + \underline{\varphi})$$

2.3.2 Überlagerung von Sinusfunktionen unterschiedlicher Frequenz ($\omega_1 \neq \omega_2$)

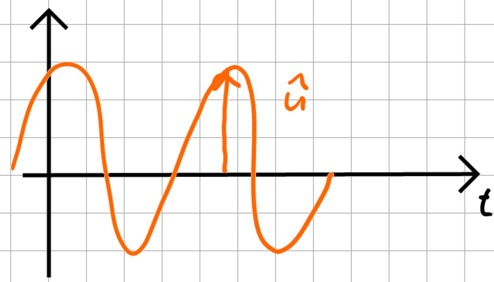


2.5 Komplexe Symbole

Motivation



$\Longrightarrow$ direkt (im Zeitbereich

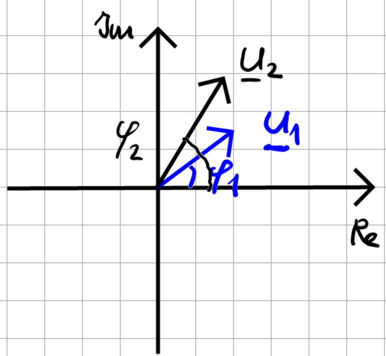


$$u(t) = u_1(t) + u_2(t)$$
$$= \hat{u} \cdot \cos(\omega t + \varphi)$$

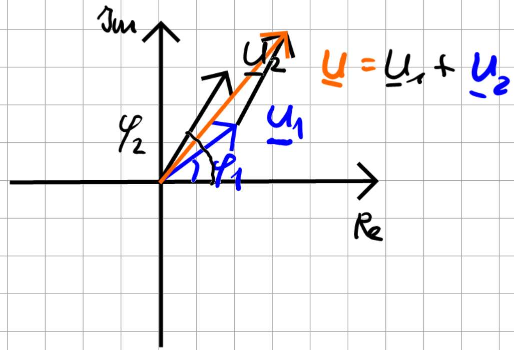
Formeln (kompliziert)

Transformation in Frequenzbereich

Rücktransformation



$\Longrightarrow$ Addition



Transformation von Zeitfunktion zum Zeiger

Vorraussetzung: ω = konstant

$$\boxed{u(t) = \hat{u} \cdot \cos(\omega t + \varphi)}$$
$$= \mathrm{Re}\{\hat{u}\cos(\omega t + \varphi) + j\,\hat{u}\sin(\omega t + \varphi)\}$$

Euler-Gleichung: $\cos x + j \sin x = e^{jx}$

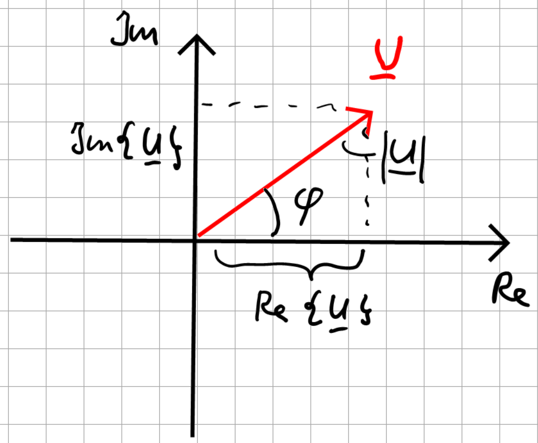
$$= \mathrm{Re}\{\hat{u}\, e^{j(\omega t + \varphi)}\}$$
$$= \mathrm{Re}\{\hat{u}\, e^{j\varphi} \cdot e^{j\omega t}\}$$

$$\boxed{\underline{U} = \frac{\hat{u}}{\sqrt{2}} \cdot e^{j\varphi}}$$

$\underline{U} = \mathrm{Re}\{\underline{U}\} + j\,\{\underline{U}\}$

$\leftarrow$ komplex

Effektivwert Winkel



Vorschrift:

$$\boxed{\begin{aligned} u(t) &= \hat{u} \cdot \cos(\omega t + \varphi) \\ &\circ\!\!-\!\!\bullet \\ \underline{U} &= \frac{\hat{u}}{\sqrt{2}} \cdot e^{j\varphi} \end{aligned}}$$

$$u_1(t) = 1\,V \cos(\omega t - 10°) \qquad \varphi$$

$$\circ\!\!-\!\!\bullet$$

$$\underline{U_1} = \frac{1V}{\sqrt{2}} \cdot e^{-j10°}$$

$$u_2(t) = 2V \cos(\omega t + 50°)$$

$$\circ\!\!-\!\!\bullet$$

$$\underline{U_2} = \frac{2V}{\sqrt{2}} \cdot e^{+j50°}$$

$$\begin{aligned}
\underline{U} &= \underline{U_1} + \underline{U_2} \\
&= \frac{1V}{\sqrt{2}} \cdot e^{-j10°} + \frac{2V}{\sqrt{2}} \cdot e^{j50°} \\
&= \frac{1V}{\sqrt{2}} \left(\cos(-10°) + j \sin(-10°)\right) + \frac{2V}{\sqrt{2}} \left(\cos(50°) + j \sin(50°)\right) \\
&= \frac{1V}{\sqrt{2}} \left(0{,}984 - j\, 0{,}174 \right) + \frac{2V}{\sqrt{2}} \left(0{,}643 + j\, 0{,}766 \right) \\
&= \frac{1}{\sqrt{2}} \left[1V \cdot \cos(-10°) + 2V \cos(50°) \right]
\end{aligned}$$

$$+ j \frac{1}{\sqrt{2}} \left[1V \cdot \sin(-10^\circ) + 2V \sin(50^\circ) \right]$$

$$\underline{U} = 1{,}605\,V + j\,0{,}96\,V$$

$$= \sqrt{(1{,}605V)^2 + (0{,}96V)^2} \cdot e^{j \arctan\left(\frac{0{,}96}{1{,}605}\right)}$$

$$\underline{U} = 1{,}87\,V \cdot e^{j31^\circ}$$

$$\bullet\!-\!\circ$$

$$u(t) = \underbrace{1{,}87V \cdot \sqrt{2}}_{\hat{u}} \cdot \cos(\omega t + 31^\circ)$$

$$= 2{,}64V \cdot \cos(\omega t + 31^\circ)$$

$$= \sqrt{2} \cdot 1{,}87V \cdot \cos(\omega t + 31^\circ)$$

Vorschrift:

$$\boxed{\begin{gathered} u(t) = \hat{u} \cdot \cos(\omega t + \varphi) \\ \circ\!-\!\bullet \\ \underline{U} = \frac{\hat{u}}{\sqrt{2}} \cdot e^{j\varphi} \end{gathered}}$$

Beispiele:

1.) $u(t) = 200V \cdot \cos(\omega t + 30^\circ)$

2.) $u(t) = 325V \cdot \cos(\omega t)$

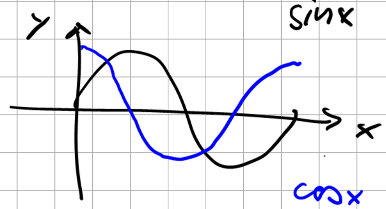
3.) $i(t) = 17A \cdot \cos(\omega t - 123°)$

4.) $i(t) = 16A \cdot \cos(\omega t - 580°)$

5.) $u(t) = -7V \cdot \cos(\omega t + 17°)$

6.) $u(t) = 23V \cdot \sin(\omega t - 200°)$

$= 23V \cos(\omega t - 200° - 90°)$



$\sin(x + 90°) = \cos x$

$\sin(x) = \cos(x - 90°)$

Lösung:

1.) $\underline{U} = \frac{200V}{\sqrt{2}} \cdot e^{j30°}$

$= 141{,}4\,V \cdot e^{j30°}$

2.) $\underline{U} = \frac{325V}{\sqrt{2}} \cdot e^{j0°}$

$= 230V$

3.) $\underline{I} = \frac{17A}{\sqrt{2}} e^{-j123°}$

$= 12A \cdot e^{-j123°}$

4.) $\underline{I} = \frac{16A}{\sqrt{2}} \cdot e^{-j580°}$

$= \frac{16A}{\sqrt{2}} \cdot e^{-j220°}$

$= 11{,}3A \cdot e^{+j140°}$

5.) $u(t) = -7A \cdot \cos(\omega t + 17°)$

$= 7A \cdot \cos(\omega t + 17° \pm 180°)$

$= 7A \cdot \cos(\omega t - 163°)$

$\underline{U} = -\frac{7A}{\sqrt{2}} \cdot e^{j17°}$

$\underline{U} = \frac{7A}{\sqrt{2}} \cdot e^{-j163°} = 4{,}95A \cdot e^{-j163°}$